Structure

1. Store and Print Student Details - Basic Structure

```
#include <stdio.h>
struct student
{
int roll;
char name[50];
int age;
float marks;
}s;
int main()
{
scanf("%d",&s.roll);
scanf(" %[^\n]",s.name);
scanf("%d",&s.age);
scanf("%f",&s.marks);
printf("Roll No: %d\n",s.roll);
printf("Name:  %s\n",s.name);
printf("Age: %d\n",s.age);
printf("Marks: %.2f",s.marks);
return 0;
}
```

2. Calculate Average Marks of Students - Array of Structures

```
#include <stdio.h>
struct student
{
int roll;
char name[50];
float marks;
}s;
int main()
{
int n;
scanf("%d",&n);
struct student s[n];
for(int i=0;i<n;i++)
{
scanf("%d",&s[i].roll);
scanf(" %[^\n]",s[i].name);
scanf("%f",&s[i].marks);
}
float aver;
for(int j=0;j<n;j++)
{
aver+=s[j].marks;
}
printf("Average Marks: %.2f",aver/n);
return 0;
}
```

3. Student Address Details - Nested Structure

```
#include <stdio.h>
struct student
{
int roll;
char name[50];
struct address
{
```

```
char city[50];
int pincode;
}add;
};
int main()
{
struct student s;
scanf("%d",&s.roll);
scanf(" %[^\n]",s.name);
scanf(" %[^\n]",s.add.city);
scanf("%d",&s.add.pincode);
printf("Roll No: %d\n",s.roll);
printf("Name: %s\n",s.name);
printf("City: %s\n",s.add.city);
printf("Pincode: %d",s.add.pincode);
return 0;
}
```

4. Update Student Marks - Pointer to Structure

```
#include <stdio.h>
#include <stdlib.h>
struct student
{
int roll;
char name[50];
float marks;
float updatedmark;
};
int main()
{
struct student *s;
s=(struct student *)malloc(sizeof(struct student ));
scanf("%d",&s->roll);
scanf(" %[^\n]",s->name);
```

```
scanf("%f",&s->marks);
scanf("%f",&s->updatedmark);
s->updatedmark+= s->marks;
if(s->updatedmark>100)
{
printf("Updated Mark: 100");
}
else
{
printf("Updated Mark: %.2f",s->updatedmark);
}
return 0;
}
```

5.Pass Structure to Function - Grade Finder

```
#include <stdio.h>
struct student
{
char name[30];
int mark;
};
void grade(char name[],int mark)
{
if(mark>=90)
{
printf("%s Grade: A",name);
}
else if(mark>=75)
{
printf("%s Grade: B",name);
}
else if(mark>=50)
{
printf("%s Grade: C",name);
```

```
}
else
{
printf("%s Grade: F",name);
}
}
int main()
{
struct student s;
scanf(" %[^\n]",s.name);
scanf("%d",&s.mark);
grade(s.name,s.mark);
return 0;
}
```

6. Return Structure from Function - Create Student Record

```
#include <stdio.h>
struct student
{
int roll;
char name[30];
float mark;
};
struct student createStudent( int roll,char name[],float mark)
{
struct student s;
s.roll=roll;
int i;
for( i=0;name[i]!='\0';i++)
```

```
{
s.name[i]=name[i];
}
s.name[i]='\0';
s.mark=mark;
return s;
}
int main()
{
struct student s;
scanf("%d",&s.roll);
scanf(" %[^\n]",s.name);
scanf("%f",&s.mark);
s=createStudent(s.roll,s.name,s.mark);
printf("Student Create \n");
printf("Roll No: %d\n",s.roll);
printf("Name: %s\n",s.name);
printf("Marks: %.2f",s.mark);
return 0;
}
```

7. Student Percentage Using typedef Structure

```
#include <stdio.h>
typedef struct
{
int roll;
```

```
char name[30];
int mark1,mark2,mark3;
}Student;
void percentage(int mark1,int mark2,int mark3)
{
float total=mark1+mark2+mark3;
float percentage=(total/300)*100;
printf("Total: %.f\n",total );
printf("Percentage: %.2f",percentage);
}
int main()
{
Student s;
scanf("%d",&s.roll);
scanf(" %[^\n]",s.name);
scanf("%d%d%d",&s.mark1,&s.mark2,&s.mark3);
percentage(s.mark1,s.mark2,s.mark3);
return 0;
}
```

8. Dynamic Student Records - malloc with Structure Array

```
#include <stdio.h>
#include <stdlib.h>
struct student
{
```

```
int roll;
char name[30];
float mark;
}s;
void studentrecords(struct student*arr,int n)
{
float total=0,average;
for(int i=0;i<n;i++)
{
total+=arr[i].mark;
}
average=total/n;
printf("Total Marks: %.2f\n",total);
printf("Average Marks:%.2f",average);
}
int main()
{
int n;
scanf("%d",&n);
struct student *arr=(struct student*)malloc(n*sizeof(struct
student));
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i].roll);
scanf(" %[^\n]",arr[i].name);
```

```
scanf("%f",&arr[i].mark);
}
studentrecords(arr,n);
return 0;
}
```

9. Student Fee Balance - Structure with Multiple Data Types

```
#include <stdio.h>
struct studentfee
{
int roll;
char name[30];
float totalfee;
double paidfee;
}s;
int main()
{
scanf("%d",&s.roll);
scanf(" %[^\n]",s.name);
scanf("%f",&s.totalfee);
scanf("%lf",&s.paidfee);
float balance =s.totalfee-s.paidfee;
printf("Balance Fee: %.2f",balance);

return 0;
```

```
}
```

10. Student Attendance Report - Array of Nested Structures

```
#include <stdio.h>
#include <stdlib.h>
struct student
{
int roll;
char name[30];
struct attendance
{
float totalclasses;
float attendedclasses;
}add;
};
int main()
{
int n;
scanf("%d",&n);
struct student *arr=(struct student*)malloc(n*sizeof(struct
student));
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i].roll);
scanf(" %[^\n]",arr[i].name);
```

```
scanf("%f",&arr[i].add.totalclasses);
scanf("%f",&arr[i].add.attendedclasses);
}float attendance=0;
for(int i=0;i<n;i++)
{
attendance=(arr[i].add.attendedclasses/arr[i].add.totalclasses)*1
00;
printf("%s Attendance :%.2f%\n",arr[i].name,attendance);
}
return 0;
}
```